

Unit.V) Decision Theory and Game Theory.

- Decision making is a process of selecting from a set of alternative courses of action, that comes, which meets the objective of decision problem, as compared to the others.

Game Theory :-

- 1) Player :- Each participant is a player.
- 2) Stratergies :- The decision rule by which a player determines his course of action during every play is called strategy.
- 3) Pure stratergy :- If a player decides to use only one particular ~~games~~ course of action during every play, he is said to use pure strategy.
- 4) Mixed stratergy :- If a player decides in advance to use all or some of his available course of action in some fixed proportion, he is said to use mixed strategy.
- 5) Pay-off :- The outcome of playing the game is called pay-off.
- 6) Two - Person zero Sum Game :- A game with two players where a gain of one player is equal to the loss of the other is known as a two-person zero sum game.
- 7) Pay-off matrix :- In a two - person zero sum game, the resulting gain represented in the form of matrix, is called pay-off matrix.

8) Maximin :- A maximum element among the minimum elements selected for each row is called maximin.

9) Minimax :- A minimum element among the maximum elements selected for each column is called minimax.

10) Stable Game :- A game is called stable game if both the players can select pure strategies such that neither player find it beneficial to alter his strategies.

Here, $\text{maximin} = \text{minimax}$.

11) Unstable game :- If a maximin is not equal to minimax then it is called as Unstable game.

12) Value of the game :- The common value of maximin and minimax of the stable game is called as stable game value of the game.

13) Saddle point :- The position in the matrix where $\text{maximin} = \text{minimax}$ is known as saddle point.

14) ~~If the~~ fair game :- If the value of the game is zero then it is called as fair game.

15) Minimax Principle :- The principle which implies that each competitor will act so as to minimize his maximum loss or maximize his minimum gain is known as minimax principle.

Player B					
Player A	I	II	III	IV	
	1	2	3	4	
	5	6	7	8	
	9	10	11	12	

Two - Person Zero Sum Game with Saddle point :-

- To obtain a soln of game, it is feasible to find out -

1) The best strategy for player A.

2) The best strategy for player B.

3) Value of the game

- Rules for determining a Saddle point :-

1) At the right of each row, write the row minimum and mark the largest among them.

2) At the bottom of each column, write the column maximum and mark the smallest among them.

3) If this mark elements are same, then the entry where the corresponding row and column intersect is called a saddle point.

4) If this two marked element are not same, then we say that there is no Saddle point.

5) If there is more than one saddle point then there will be more than one solution, each solution corresponding to each saddle point.

Q. Determine the saddle point and optimum strategies for each player. comment at the value of the game.

		player B.		
		B ₁	B ₂	B ₃
player A	A ₁	-3	-2	0
	A ₂	2	0	2
	A ₃	5	-2	4

Soln:- Given pay-off matrix is,

		Player B.			
		B ₁	B ₂	B ₃	Row min
Player A	A ₁	-3	-2	0	-3
	A ₂	2	0	2	0 ← minimax, maximin.
	A ₃	5	-2	4	-2
Column min max		5	0 ↑ minimax.	4	

Here, $\text{maximin} = \text{minimax} = 0$
 Value Saddle point is at (2,2).

Value of the game is 0.

∴ Above game is fair game
 and optimal strategy for player B is B₂.

————— || ————— A₂ is A₂.

2) Find minimax and maximin for the following matrix.

$$\begin{bmatrix} -2 & -1 & 5 \\ 2 & 1 & 3 \\ 4 & 2 & -3 \end{bmatrix}$$

soln:- Here, pay-off matrix is,

		Row min		
		-2	-1	5
		2	1	3
		4	2	-3
		-2	1 ← maximin	-3
Column max		4	2 ↑ minimax.	5

Here, $\text{maximin} = 1$, $\text{minimax} = 2$.

∴ Above matrix does not have Saddle point

∴ Above game is unstable game

3) Determine the value of λ in following game such that the game is strictly determinable.

		player B		
		B ₁	B ₂	B ₃
player A	A ₁	λ	6	2
	A ₂	-1	λ	$-\lambda$
	A ₃	-2	4	λ

Soln: - firstly, ignore the value of λ , find the maximin and minimax of the given pay-off matrix.

		player B			
		B ₁	B ₂	B ₃	Row min
player A	A ₁	λ	6	2	$\boxed{2}$ ← maximin
	A ₂	-1	λ	$-\lambda$	$-\lambda$
	A ₃	-2	4	λ	-2

column
max

$\boxed{-1}$
↑

minimax

Here,

$$\text{maximin} = 2 \text{ and } \text{minimax} = -1$$

$$\therefore -1 \leq \lambda \leq 2$$

$\therefore$ Above game is strictly determinable.

		B ₁	B ₂	B ₃
player A	A ₁	2	-1	-2
	A ₂	1	1	0
	A ₃	-1	2	4

$\boxed{1}$ ← maximin
 ↑
 minimax

Two-person zero Sum game without Saddle point :-

1) Algebraic Method for Solving 2x2 Game without Saddle point :-

• Working Rule :-

1) Calculate each row difference.

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \begin{array}{l} \text{Row difference} \\ a_{11} - a_{12} \\ a_{21} - a_{22} \end{array}$$

2) Calculate sum of two row difference. Divide each row difference by this sum to get ratios.

3) Interchange the ratios to get probabilities.

4) Calculate the column differences.

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \begin{array}{l} \text{Column difference} \\ a_{11} - a_{21} \\ a_{12} - a_{22} \end{array}$$

5) Calculate the sum of two column difference, Divide each column difference by their sum to get the ratios.

6) Interchange the ratios to get probabilities.

Q. Solve the following game.

		I	II
player A	I	20	-6
	II	-4	3

Soln:- Here,

		I	II	
max	I	20	-6	-6
	II	-4	3	-4
		20	3	

← maximin
← minimax

Here, maximin $\neq$ minimax.

Hence given problem does not have saddle point.

So, we will solve it by using Algebraic method.

		player B.		Row diff	Ratio's	probabilities
		I	II			
player A	I	20	-6	26	26/33	7/33
	II	-4	3	7	7/33	26/33
column difference		24	9			
Ratio's		24/33	9/33			
probabilities		9/33	24/33			

$\therefore$ Optimal strategies for player A are $S_A = \{7/33, 26/33\}$

_____||_____ B are $S_B = \{9/33, 24/33\}$.

So, here, $x_1 = \frac{7}{33}$, $x_2 = \frac{26}{33}$

Hence, the value of the game $= a_{11}x_1 + a_{21}x_2$

$$= 20 \times \frac{7}{33} + (-4) \times \frac{26}{33}$$

$$= \frac{36}{33}$$

2) Solve the following two persons 2×2 mixed strategy game.

		player B.	
		I	II
player A	I	10	4
	II	7	8

Soln:- we solve this by Algebraic method.

Since there is no Saddle point.

player B.

		player B.		Row diff	Ratio	probabilities
		I	II			
player A	I	10	4	6	$\frac{6}{7}$	$\frac{1}{7}$
	II	7	8	1	$\frac{1}{7}$	$\frac{6}{7}$
Column diff		3	4	7		
Ratio		$\frac{3}{7}$	$\frac{4}{7}$			
probability		$\frac{4}{7}$	$\frac{3}{7}$			

∴ Optimal strategies for player A are $S_A = \left\{ \frac{1}{7}, \frac{6}{7} \right\}$

for player B are $S_B = \left\{ \frac{4}{7}, \frac{3}{7} \right\}$

∴ Value of the game $= a_{11}x_1 + a_{21}x_2$

where $x_1 = \frac{1}{7}, x_2 = \frac{6}{7}$

∴ Value of the game $= 10 \times \frac{1}{7} + 7 \times \frac{6}{7}$

$= \frac{52}{7}$

Dominance Method (Solution $m \times n$ Game without Saddle Point) :-

• Dominance Principle :-

If no pure strategies exist, then by eliminating certain strategies (row or columns) by dominance, we can solve the given game by some mixed strategy.

* Rules of Dominance :-

- 1) If every entry in the i th row is less than or equal to the corresponding entry in j th row, then we eliminate the i th row.
- 2) If every entry in the j th column is less than or equal to corresponding entry in i th column, then we eliminate j th column.
- 3) If average of two or more rows dominates a row, then eliminate the dominated row.
- 4) If average of two or more columns dominated by a column then eliminate the column.
- 5) The resulting pay-off matrix is either of order 2×2 or $2 \times n$ or $m \times 2$.
- 6) If payoff matrix is of order 2×2 then solve it by dominance Algebraic method.
Also, if payoff matrix is of order $2 \times n$ or $m \times 2$, then solve it by graphical method.

Example :- 1) Reduce the following game by the dominance principal and find optimal solⁿ.

		player B		
		B ₁	B ₂	B ₃
player A	A ₁	12	10	4
	A ₂	7	5	2
	A ₃	9	7	8

Solⁿ:- We reduce the above game by using dominance rule

Step: I) Compare the A_2 and A_3 row, all elements of A_2 are less than the corresponding elements of A_3 . So, we delete A_2 row from the given pay-off matrix.

The Reduce payoff matrix as:

	B_1	B_2	B_3
A_1	12	10	4
A_3	9	7	8

Step: II) Compare the B_1 and B_2 columns, all elements of B_1 column are greater than the corresponding elements of B_2 column. So, we delete B_1 column from the given payoff matrix.

	B_2	B_3
A_1	10	4
A_3	7	8

Step: III) Now, we solve above pay-off matrix by using Algebraic method.

	Player B		Row diff.	Ratio	Probabilities
	B_2	B_3			
A_1	10	4	6	$6/7$	$1/7$
A_3	7	8	1	$1/7$	$6/7$
column diff	3	4	7		
Ratio	$3/7$	$4/7$			
Probabilities	$4/7$	$3/7$			

Here, optimal strategy for player A = $\left\{ \frac{1}{7}, \frac{6}{7} \right\}$

and Optimal strategy for player B = $\left\{ \frac{4}{7}, \frac{3}{7} \right\}$.

$$\text{Value of game} = 10 \times \frac{1}{7} + 7 \times \frac{6}{7} = \frac{52}{7}$$

2) Reduce the following game by dominance principle and solve it.

		player B				
		I	II	III	IV	V
player A	I	1	3	2	7	4
	II	3	4	1	5	6
	III	6	5	7	6	5
	IV	2	0	6	3	1

Soln:- We reduce the size of given pay-off matrix by dominance principle.

Here, row IV is dominated by row III.
 $\therefore$ Deleting IV row, we get

		player B				
		I	II	III	IV	V
player A	I	1	3	2	7	4
	II	3	4	1	5	6
	III	6	5	7	6	5

Now, column IV is dominated by average of column I, and II, also column V is dominated by column II. Therefore, delete column IV and V.

		player B		
		I	II	III
player A	I	1	3	2
	II	3	4	1
	III	6	5	7

In above matrix, row I and row II is dominated by row III. Therefore, we delete row I and II, we get reduced matrix as,

		Player B		
		I	II	III
Player A	III	6	5	7

Here, column II is dominated by column I and column III, we have,

		Player B	
		II	
Player A	III	5	

∴ Optimal strategy for player A is III and for player B is II.

∴ Value

① Explain the following terms.

① Fair Game :- If the value of the game is zero, then it is called as fair game.

② ^{Mixed} Pure strategy :- If a player (or competitor) decides in advance to use all or some of his available course of action in some fixed proportion, then he is said to use mixed strategy.

③ Pure strategy :- If a player decides to use only one particular course of action during play, he is said to use a pure strategy.